

A Full Bayesian State-Space and Belief-State Control Specification for Credit-Card Customer NPV Decisioning

Chak Wong

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Abstract

This note gives a full stochastic-process assembly of the credit-card customer NPV proposal into a constrained Bayesian state-space model with belief-state control. The goal is not to replace the proposal’s valuation semantics. It is to make the dynamic object explicit enough that forward simulation under a fixed declared policy is mathematically well defined and later offline policy evaluation or constrained challenger optimization can be discussed without handwaving. The note reuses the current proposal’s valuation object, wallet accounting, module decomposition, path-engine semantics, and policy-value conventions, then adds the missing layers required for a committee-level stochastic specification: timing and filtration, explicit latent and observed state blocks, module-level controlled transition equations, observation equations, Bayesian filtering recursions, constrained Bellman equations, and an identification / governance boundary. The resulting object is a lifecycle belief-state control model, not a claim that the bank is already ready for production reinforcement learning. The literature supports the sequential control framing, partial observability, state-space filtering, and offline policy-evaluation toolkit [4, 6, 11, 12, 14, 15, 17, 19–21, 23, 24]; it does not identify this issuer’s exact latent state, reward ledger, constraint set, or safe rollout threshold.

1 Introduction and contribution boundary

The credit-card NPV proposal already contains the essential economic pieces of a dynamic decision system: a versioned valuation object, a modular state structure, wallet accounting, action-dependent terms, a path engine, a cash-flow compiler, and policy-value notation. What it does not yet present is a single stochastic process that assembles those pieces into one mathematically explicit control object. Without that object the proposal can say that future policy matters, but it cannot fully specify what is being filtered, what is transitioning, what is observed with noise or lag, or exactly which state a policy should condition on.

That gap matters because many committee-level objections are structural rather than stylistic. If the proposal names outside wallet, movable wallet, repayment propensity, rewards response, attrition, PD/LGD/EAD, and macro sensitivity as important drivers, then a mathematically serious note must say where those objects live in the state, how they evolve, how they are observed, and how they enter the reward and feasible action set. A high-level POMDP discussion is not enough.

This note therefore contributes a full stochastic-process specification. It makes five bounded claims.

1. Credit-card customer value can be represented as a sequential control problem, with discounted incremental NPV as the policy-value object [19, 20, 23].

2. A fully observed MDP is generally an approximation because important economic drivers are latent or only partially observed; a belief-state or POMDP formulation is therefore the correct umbrella language [14, 21].
3. The current proposal’s macro, spend, balance, payment, loss, reward, attrition, and relationship modules can be assembled into one controlled state-space system.
4. Bayesian filtering supplies the correct updating object: the policy should depend on the posterior distribution of the latent state, not only on a point estimate [4, 6, 24].
5. Offline policy evaluation and safe-improvement tools are relevant only after the dynamic system, logged behavior policy, and support conditions are explicit [11, 12, 15, 17].

The note does *not* claim that the literature already identifies this issuer’s exact latent state vector, its competitor-wallet dynamics, its future policy bundle, or its live deployment threshold. Those remain bank-specific empirical and governance questions.

2 Notation and timing contract

This note reuses the main proposal’s global notation. Unit i is a customer, account, prospect, application, or account-action unit. Time t is monthly. Action is a . The decision-specific baseline is $a_0(d)$. Decision context is d . Scenario and valuation bundle is s . Downstream account-management policy bundle is π^{down} . Governance and hard admissibility are represented by π^{gov} . The decision-time information set is $\mathcal{I}_{id}$ or X_{it}^d .

The current proposal’s one-period incremental cash-flow primitive and NPV functional are retained:

$$\begin{aligned} \Delta CF_{i,t+h}(a, \pi; s) = & \Delta PPNR_{i,t+h}(a, \pi; s) - \Delta EL_{i,t+h}(a, \pi; s) - \Delta Kchg_{i,t+h}(a, \pi; s) \\ & - \Delta Tax_{i,t+h}(a, \pi; s) + \Delta RelValue_{i,t+h}(a, \pi; s), \end{aligned} \quad (1)$$

$$\Delta NPV_i(a; d, s, \pi) = -C_i^{acq}(a) + \mathbb{E} \left[\sum_{h=0}^H \delta_h \Delta CF_{i,t+h}(a, \pi; s) + \delta_H \Delta TV_{i,t+H}(a, \pi; s) \mid X_{it}^d \right]. \quad (2)$$

This note distinguishes three policy objects. The symbol π^{down} denotes the downstream operating policy bundle assumed in valuation. The symbol μ denotes the historical logged behavior policy that generated observed actions. The symbol φ denotes a candidate or challenger rule to be simulated, evaluated, or eventually optimized. When the older proposal notation uses π generically, the reader should check whether it refers to π^{down} , μ , φ , or the governance bundle π^{gov} .

2.1 Intra-month timing

A committee-level dynamic model needs an explicit within-month sequence. Let the month- t cycle be:

- T1.** At the opening of month t , the bank has observed history H_{it}^- and a posterior belief b_{it}^- over the latent state.
- T2.** Macro or scenario inputs available at decision time are incorporated; lagged or not-yet-released macro variables remain unavailable.

T3. The bank constructs the decision feature state O_{it} and chooses an admissible action

$$a_{it} \in \mathcal{A}_t(O_{it}, b_{it}^-; d, s, \pi^{gov}).$$

T4. Conditional on a_{it} and the current latent/full state, within-month customer behavior realizes: activation or use, category spend, cash-advance or balance-transfer use, balances, payments, delinquency migration, rewards accrual, attrition or retention outcomes, and any collections response.

T5. Raw signals $Y_{i,t+1}$ are observed with the data lags, censoring, or partial observability implied by the source system.

T6. The posterior is updated to $b_{i,t+1}^-$ using Bayes' rule, and the month- t reward is compiled from the realized or expected incremental cash-flow mapping.

The relevant filtration should be written in two steps. The pre-action information set is

$$\mathcal{F}_{it}^- = \sigma(H_{it}^-, O_{it}, b_{it}^-), \quad (3)$$

and the action must be measurable with respect to it:

$$a_{it} = \varphi_t(O_{it}, b_{it}^-, d, s, \pi^{down}) \quad \text{or} \quad a_{it} \sim \varphi_t(\cdot \mid O_{it}, b_{it}^-, d, s, \pi^{down}). \quad (4)$$

After the action and raw signals are realized, the post-observation information set is

$$\mathcal{F}_{i,t+1}^+ = \sigma(\mathcal{F}_{it}^-, a_{it}, Y_{i,t+1}, O_{i,t+1}). \quad (5)$$

3 State decomposition

3.1 Observed state

Let the observed decision state be

$$O_{it} = (G_{it}, B_{it}, L_{it}, P_{it}, D_{it}, Rwd_{it}, Rel_{it}^{obs}, M_{it}^{obs}, C_{it}), \quad (6)$$

where:

- G_{it} is lifecycle regime,
- B_{it} is the balance-subledger vector,
- L_{it} is line and term state,
- P_{it} is observed payment behavior,
- D_{it} is observed delinquency/default/cure status,
- Rwd_{it} is observed reward and promotion state,
- Rel_{it}^{obs} is observed relationship state,
- M_{it}^{obs} is the observed macro signal vector available at decision time, and
- C_{it} is decision-context metadata such as channel, offer family, feature-cut timestamp, and eligibility indicators.

The balance-subledger vector should be explicit:

$$B_{it} = (B_{it}^{pur}, B_{it}^{cash}, B_{it}^{BT}, B_{it}^{fee}, B_{it}^{int}, B_{it}^{chg}), \quad (7)$$

so that principal, non-principal, promo, and charge-related balances are not collapsed into one scalar.

The lifecycle regime should also be explicit:

$$G_{it} \in \{\text{prospect, targeted, applicant, approved, booked, activated, active-transactor, active-revolver, promo, dormant, retained, converted, delinquent, collections, closed, charged-off}\}. \quad (8)$$

3.2 Latent state

Let the latent economic state be

$$U_{it} = (M_{it}^{lat}, W_{it}^{out}, \Theta_{it}^{route}, \Theta_{it}^{repay}, \Theta_{it}^{attr}, \Theta_{it}^{promo}, Rel_{it}^{lat}), \quad (9)$$

where:

- M_{it}^{lat} is a latent macro or local economic regime, if the bank chooses to smooth or nowcast macro conditions beyond raw releases,
- W_{it}^{out} is outside wallet or outside balance opportunity,
- Θ_{it}^{route} captures latent routing / card-on-file / issuer- preference frictions,
- Θ_{it}^{repay} captures latent repayment, liquidity, and distress state not fully summarized by observed balances and payments,
- Θ_{it}^{attr} captures latent attrition / dormancy / retention propensity,
- Θ_{it}^{promo} captures latent promotion-response and decay state, and
- Rel_{it}^{lat} captures latent relationship spillover state.

The full economic state is

$$X_{it} = (O_{it}, U_{it}). \quad (10)$$

3.3 Belief state

Because U_{it} is not observed, the control object is the posterior belief state

$$b_{it}^-(u) = p(U_{it} = u \mid H_{it}^-, s, \pi^{gov}). \quad (11)$$

The policy may use a finite-dimensional approximation to b_{it} in implementation, but the conceptual sufficient statistic is the posterior distribution itself. For readability, later equations usually write b_{it} or simply b for this pre-action belief unless the distinction between pre-action and post-observation beliefs is material.

For a first implementation, the operational belief can be a finite-dimensional summary,

$$\hat{b}_{it} = (\hat{p}_{it}^{primary}, \hat{W}_{it}^{out}, \hat{p}_{it}^{promo}, \hat{p}_{it}^{revolver}, \hat{p}_{it}^{stress}, \hat{p}_{it}^{attr}, \hat{M}_{it}^{lat}, \hat{Rel}_{it}^{lat}), \quad (12)$$

with uncertainty and evidence grade attached to entries that materially affect NPV. This is an implementable approximation to the belief-state object, not a claim that the coordinates are uniquely correct.

4 Controlled stochastic process

4.1 Action space by regime

The feasible action set must depend on lifecycle regime and governance state:

$$a_{it} \in \mathcal{A}_t(O_{it}, b_{it}; d, s, \pi^{gov}). \quad (13)$$

Representative stage-specific actions are:

- acquisition: suppress/contact, channel, offer family,
- underwriting or line assignment: approve/decline, line bucket, APR and fee terms, promo cell,
- existing-account treatment: line change, repricing, promo, retention, conversion,
- collections: no treatment, reminder, hardship, plan, escalation.

Product terms should be treated as a governed action vector,

$$a_{it} = (\ell_{it}, r_{it}^{pur}, r_{it}^{cash}, r_{it}^{BT}, r_{it}^{promo}, T_{it}^{promo}, f_{it}, q_{it}, b_{it}^{signup}, d_{it}^{disc}), \quad (14)$$

consistent with the main proposal's offer-vector representation.

4.2 Macro block

A full stochastic note should include an explicit macro law of motion. One flexible specification is a linear-Gaussian latent macro regime with observed releases:

$$M_{i,t+1}^{lat} = A_M M_{it}^{lat} + \eta_{t+1}^M, \quad \eta_{t+1}^M \sim \mathcal{N}(0, Q_M), \quad (15)$$

$$M_{i,t+1}^{obs} = H_M M_{i,t+1}^{lat} + \nu_{t+1}^M, \quad \nu_{t+1}^M \sim \mathcal{N}(0, R_M), \quad (16)$$

with release lags handled through missing or stale components of $M_{i,t+1}^{obs}$.

If the bank prefers deterministic stress scenarios, then (15) can be replaced by an exogenous scenario path. But for a committee-level stochastic-process note, an explicit macro block is the cleanest baseline because the main proposal repeatedly makes macro state a structural driver of spend, payment, loss, and attrition [1, 9, 10, 13].

4.3 Usage, wallet, and routing block

Let total cardable category- j spending opportunity satisfy

$$\log(1 + C_{ij,t+1}) = \mu_{ij}^C + \beta_j^{M\top} M_{i,t+1}^{lat} + \beta_j^{X\top} X_{it}^{obs} + \beta_j^{U\top} \Theta_{it}^{repay} + \varepsilon_{ij,t+1}^C, \quad (17)$$

where $C_{ij,t+1}$ is latent total cardable spend opportunity in category j . This lifts the proposal's category-spend specification into the latent dynamic system.

Let routing shares across payment rails be

$$q_{ij,t+1}(a) = \text{softmax}(\alpha_j + \Gamma_j O_{it} + \Lambda_j U_{it} + \Psi_j a_{it}), \quad (18)$$

with components corresponding to new-bank card, old-bank card, outside-card, debit/deposit, and cash/other rails. Then observed or latent rail-specific spend is

$$S_{ij,t+1}^{rail}(a) = q_{ij,t+1}^{rail}(a) C_{ij,t+1}. \quad (19)$$

This gives a true stochastic-process interpretation to the main proposal's wallet accounting and spend-source decomposition.

The outside-wallet state evolves as

$$W_{i,t+1}^{out} = h_W(W_{it}^{out}, M_{i,t+1}^{lat}, G_{it}, \Theta_{it}^{route}, a_{it}) + \varepsilon_{i,t+1}^W, \quad (20)$$

and the offer-specific movable-wallet state is a derived functional

$$W_{it}^{mov}(a) = m(U_{it}, O_{it}, a, d, s, \pi). \quad (21)$$

4.4 Lifecycle and activation block

The lifecycle state transition is controlled and stage-dependent:

$$\Pr(G_{i,t+1} = g' \mid G_{it} = g, O_{it}, U_{it}, a_{it}, s) = p_G(g' \mid g, O_{it}, U_{it}, a_{it}, s). \quad (22)$$

For adoption and first use, a useful two-step structure is

$$A_{im,t+1}^* = \alpha_m^\top O_{it} + \gamma_m^\top U_{it} + \xi_{im,t+1}, \quad A_{im,t+1} = \mathbf{1}\{A_{im,t+1}^* > 0\}, \quad (23)$$

$$U_{ijm,t+1}^{use} = \beta_m^\top O_{it} + \phi_m A_{im,t+1} + \rho_m^\top U_{it} + \epsilon_{ijm,t+1}, \quad (24)$$

with the new-bank-card rail choice generated through the routing system in (18). This retains the proposal's adoption/use logic while placing it inside a monthly controlled process.

4.5 Balance, payment, and line block

The stock-flow identity must hold pathwise:

$$B_{i,t+1}^{pur} = B_{it}^{pur} + S_{i,t+1}^{newbank,pur} - Pay_{i,t+1}^{pur} - Chg_{i,t+1}^{pur}, \quad (25)$$

$$B_{i,t+1}^{cash} = B_{it}^{cash} + S_{i,t+1}^{cashadv} - Pay_{i,t+1}^{cash} - Chg_{i,t+1}^{cash}, \quad (26)$$

$$B_{i,t+1}^{BT} = B_{it}^{BT} + BT_{i,t+1} - Pay_{i,t+1}^{BT} - Chg_{i,t+1}^{BT}, \quad (27)$$

$$B_{i,t+1}^{fee} = B_{it}^{fee} + Fee_{i,t+1} - Pay_{i,t+1}^{fee}, \quad (28)$$

$$B_{i,t+1}^{int} = B_{it}^{int} + Int_{i,t+1} - Pay_{i,t+1}^{int}, \quad (29)$$

$$B_{i,t+1}^{tot} = B_{i,t+1}^{pur} + B_{i,t+1}^{cash} + B_{i,t+1}^{BT} + B_{i,t+1}^{fee} + B_{i,t+1}^{int}, \quad (30)$$

where the flows indexed by $t+1$ are realized during $(t, t+1]$ after action a_{it} is chosen. The interest, fee, and charge-off components are generated by the action, subledger, and delinquency states.

Payment behavior should be controlled by latent repayment state and observed balance burden:

$$Pay_{i,t+1} = g_P(B_{it}^{tot}, Util_{it}, G_{it}, D_{it}, O_{it}, \Theta_{it}^{repay}, a_{it}, M_{i,t+1}^{lat}) + \varepsilon_{i,t+1}^P. \quad (31)$$

Line and term dynamics are also controlled:

$$L_{i,t+1} = g_L(L_{it}, B_{it}^{tot}, D_{it}, O_{it}, a_{it}, \pi^{gov}) + \varepsilon_{i,t+1}^L. \quad (32)$$

4.6 Risk, delinquency, and recovery block

The delinquency/default process is a controlled finite-state Markov chain or hazard model. One flexible representation is

$$\Pr(D_{i,t+1} = d' \mid D_{it} = d, G_{it}, B_{it}^{tot}, Pay_{i,t+1}, L_{it}, O_{it}, U_{it}, a_{it}, M_{i,t+1}^{lat}) = p_D(d' \mid \cdot). \quad (33)$$

Expected exposure at default is tied to the subledgers and undrawn behavior:

$$EAD_{i,t+1} = B_{i,t+1}^{pur} + B_{i,t+1}^{cash} + B_{i,t+1}^{BT} + B_{i,t+1}^{fee} + B_{i,t+1}^{int} + UndrawnDraw_{i,t+1}. \quad (34)$$

LGD and recovery evolve conditionally on default state and collections action:

$$Rec_{i,t+1:t+W}^{work} = g_R(D_{i,t+1}, O_{it}, U_{it}, a_{it}, M_{i,t+1}^{lat}) + \varepsilon_{i,t+1}^R, \quad (35)$$

$$LGD_{i,t+1} = 1 - \frac{\delta^{work} Rec_{i,t+1:t+W}^{work}}{\max(EAD_{i,t+1}, \varepsilon_{EAD})}. \quad (36)$$

Here $Rec_{i,t+1:t+W}^{work}$ is discounted recovery over the workout window, conditional on default or charge-off. In implementation, LGD is defined only on the default/exposure population and must reconcile to approved recovery and charge-off conventions.

These equations make the main proposal's PD/LGD/EAD structure part of the full state process rather than a detached risk appendix.

4.7 Rewards, promotions, attrition, conversion, and relationship block

Promotion state should evolve explicitly:

$$Rwd_{i,t+1} = g_{Rwd}(Rwd_{it}, S_{i,t+1}^{newbank}, a_{it}, \Theta_{it}^{promo}) + \varepsilon_{i,t+1}^{Rwd}, \quad (37)$$

with latent promo response state

$$\Theta_{i,t+1}^{promo} = A_{\Theta,p} \Theta_{it}^{promo} + \Gamma_p a_{it} + \eta_{i,t+1}^p. \quad (38)$$

Attrition and dormancy should also be explicit:

$$\Pr(G_{i,t+1} \in \{\text{dormant}, \text{closed}\} \mid G_{it}, O_{it}, U_{it}, S_{it}, B_{it}, a_{it}) = p_A(\cdot \mid \cdot), \quad (39)$$

and product conversion should satisfy

$$\Pr(Product_{i,t+1} = p' \mid Product_{it} = p, G_{it}, O_{it}, U_{it}, a_{it}) = p_{conv}(p' \mid p, \cdot). \quad (40)$$

Finally, relationship spillover should have a dynamic block rather than appear only as a terminal scalar:

$$Rel_{i,t+1}^{lat} = g_{Rel}(Rel_{it}^{lat}, Rel_{it}^{obs}, a_{it}, G_{it}, M_{i,t+1}^{lat}) + \varepsilon_{i,t+1}^{Rel}. \quad (41)$$

5 Observation system

A full state-space note must say what the bank observes and how. The observation system is module-specific.

5.1 Transactional and statement observations

Observed new-bank-card and old-bank-card spend are exact or nearly exact ledger aggregates conditional on source-system quality:

$$Y_{ij,t+1}^{new} = S_{ij,t+1}^{newbank} + \nu_{ij,t+1}^{new}, \quad (42)$$

$$Y_{ij,t+1}^{old} = S_{ij,t+1}^{oldbank} + \nu_{ij,t+1}^{old}, \quad (43)$$

with ν typically zero or small reconciliation noise. Statement balances, payments, fees, and interest are observed through the ledger:

$$Y_{i,t+1}^B = B_{i,t+1}^{tot} + \nu_{i,t+1}^B, \quad (44)$$

$$Y_{i,t+1}^{Pay} = Pay_{i,t+1} + \nu_{i,t+1}^{Pay}. \quad (45)$$

5.2 Risk and bureau observations

Delinquency and charge-off are observed discrete states,

$$Y_{i,t+1}^D = h_D(D_{i,t+1}) + \nu_{i,t+1}^D, \quad (46)$$

while bureau refreshes are periodic and noisy observations of broader external credit condition,

$$Y_{i,t+1}^{bureau} = h_{bureau}(U_{i,t+1}, D_{i,t+1}, B_{i,t+1}^{tot}) + \nu_{i,t+1}^{bureau}. \quad (47)$$

5.3 Wallet and routing anchors

The bank does not directly observe outside wallet every month. It may observe partial anchors such as balance transfers into the card, payoff events, account-aggregation samples, network aggregates, or deposit-rail substitution signals. Represent these as sparse or partial measurements:

$$Y_{i,t+1}^{wallet} = h_{wallet}(W_{i,t+1}^{out}, \Theta_{i,t+1}^{route}, O_{i,t+1}) + \nu_{i,t+1}^{wallet}, \quad (48)$$

where missingness itself is informative and must be retained in the filter.

5.4 Macro observations

Observed macro data are release-lagged measurements of the latent macro regime, already written in (16). The note should treat revised macro data as different objects from the as-of-decision releases used in the filter.

6 Bayesian filtering and belief-state recursion

The full posterior satisfies

$$b_{i,t+1}(u') \propto p(Y_{i,t+1}, O_{i,t+1} \mid u', O_{it}, a_{it}, s, \pi^{down}) \int p(u' \mid u, O_{it}, a_{it}, s, \pi^{down}) b_{it}(u) du. \quad (49)$$

where $O_{i,t+1}$ is the approved next decision state constructed from raw signals. If that construction is deterministic, the likelihood includes the state-construction rule rather than an additional random observation. For the full enterprise model, exact filtering is generally infeasible. The proposal should therefore define a blockwise filtering architecture.

Macro block. If (15)–(16) is used, the macro posterior can be updated with Kalman-style state-space recursions [6].

Lifecycle and delinquency blocks. Finite-state transitions for G_t and D_t can use exact hidden Markov model recursions conditional on observed transactional and delinquency signals.

Latent repayment and promo response blocks. If these are modeled as approximately Gaussian dynamic linear states, posterior updates can use dynamic-linear-model recursions [24].

Outside-wallet and routing block. Because the observation system is sparse and nonlinear, this block should be filtered with particle or sequential Monte Carlo approximations when the bank requires full posterior uncertainty [4].

The implementation may retain a structured approximate posterior,

$$b_{it}(u) \approx b_{it}^M(M^{lat})b_{it}^{wallet}(W^{out}, \Theta^{route})b_{it}^{repay}(\Theta^{repay})b_{it}^{promo}(\Theta^{promo})b_{it}^{rel}(Rel^{lat}), \quad (50)$$

provided the approximation error is treated as part of model risk rather than ignored.

Proposition 6.1 (Belief-state sufficiency). Under the controlled Markov transition and observation equations above, any history-dependent policy has an equivalent policy measurable with respect to the current observed state O_{it} and belief state b_{it} , with the same finite-horizon value. The control problem is therefore an MDP on belief space [14, 21].

Proof. By the tower property, any continuation value depends on past history only through the conditional distribution of the latent state and the observed state that indexes the transition, reward, and feasible-action correspondence. The filter in (49) constructs exactly that conditional distribution. Replacing a history-dependent policy by one measurable with respect to (b_{it}, O_{it}) therefore preserves the distribution of future rewards and the resulting value. $\square$

7 Reward and valuation mapping

The reward must be compiled from the same economic decomposition already proposed, not replaced by an abstract utility index. Conditional on the full state, action, next state, and realized monthly signals,

$$\begin{aligned} r_{i,t+1}(X_{it}, a_{it}, X_{i,t+1}, Y_{i,t+1}; s, \pi^{down}) = & \Delta PPNR_{i,t+1}(X_{it}, a_{it}, X_{i,t+1}; s, \pi^{down}) \\ & - \Delta EL_{i,t+1}(X_{it}, a_{it}, X_{i,t+1}; s, \pi^{down}) \\ & - \Delta Kchg_{i,t+1}(X_{it}, a_{it}, X_{i,t+1}; s, \pi^{down}) \\ & - \Delta Tax_{i,t+1}(X_{it}, a_{it}, X_{i,t+1}; s, \pi^{down}) \\ & + \Delta RelValue_{i,t+1}(X_{it}, a_{it}, X_{i,t+1}; s, \pi^{down}). \end{aligned} \quad (51)$$

The belief-state reward is then

$$\bar{r}_{it}(b_{it}, O_{it}, a_{it}; s, \pi^{down}) = \mathbb{E}\left[r_{i,t+1}(X_{it}, a_{it}, X_{i,t+1}, Y_{i,t+1}; s, \pi^{down}) \mid b_{it}, O_{it}, a_{it}, s, \pi^{down}\right]. \quad (52)$$

The path engine in the main proposal is then reinterpreted as simulation of the state-space system under candidate action and baseline action, with the subledger identities enforced exactly and the reward compiled from the simulated paths.

8 Constrained policy value and Bellman recursion

Let H denote the terminal valuation month for this note. The fixed-policy value under a candidate rule φ and downstream policy bundle π^{down} is

$$V_t^{\varphi; \pi^{down}}(b_{it}, O_{it}; s) = \mathbb{E}^{\varphi, \pi^{down}} \left[\sum_{h=t}^{H-1} \delta^{h-t} \bar{r}_h(b_{ih}, O_{ih}, a_{ih}; s, \pi^{down}) + \delta^{H-t} \Delta T V_{iH}(b_{iH}, O_{iH}; s, \pi^{down}) \mid b_{it}, O_{it} \right]. \quad (53)$$

The constrained Bellman recursion is

$$V_t^*(b, O; s) = \max_{a \in \mathcal{A}_t(b, O; d, s, \pi^{gov})} \left\{ \bar{r}_t(b, O, a; s, \pi^{down}) + \delta \mathbb{E} \left[V_{t+1}^*(b', O'; s) \mid b, O, a, s, \pi^{down} \right] \right\} \quad (54)$$

with pointwise hard admissibility

$$a \in \mathcal{A}_t^{hard}(O, b, d, s, \pi^{gov}), \quad (55)$$

and cumulative constraint costs

$$J_{c,k}^{\varphi}(b, O; s) = \mathbb{E}^{\varphi, \pi^{down}} \left[\sum_{h=t}^{H-1} \delta^{h-t} c_{k,h}(b_{ih}, O_{ih}, a_{ih}; s) \mid b_{it} = b, O_{it} = O \right] \leq \kappa_k. \quad (56)$$

Loss, capital, liquidity, operational, complaint, and fairness constraints are usually cumulative or portfolio-level objects, not merely one-period inequalities. They can be handled through remaining-budget state augmentation or Lagrangian relaxations when the policy class and evidence support optimization.

A fully observed MDP is the special case in which U_{it} is either empty or exactly observed and b_{it} degenerates to a point mass.

9 Module-to-process mapping

10 Identification, estimation, and evidence discipline

The stochastic-process note should classify each equation or block as one of the following:

1. accounting identity,
2. probabilistic factorization,
3. literature-supported control or filtering principle,
4. bank-specific empirical object,
5. governance or admissibility convention.

The literature directly supports:

- the MDP / dynamic treatment regime framing [19, 20];

Table 1: How the existing proposal modules enter the full stochastic process.

Module	Transition role	Observation role	Reward / constraint role
Macro and labor state	latent and observed macro regime transitions	release-lagged macro signals	scenario conditioning; stress and drift constraints
Adoption / activation / use	lifecycle transitions and route choice	application, booking, activation, first use, repeat use	gates future value and contact-policy feasibility
Wallet and cannibalization	outside wallet and routing dynamics	partial anchors from bank book, bureau, network, or experiments	determines all-bank incremental spend versus raw new-card spend
Balances / payments / lines	subledger and payment transitions	statement balances, payments, interest, fees	finance charges, funding, EAD, line constraints
Delinquency / losses / recoveries	delinquency, default, cure, and recovery transitions	delinquency buckets, charge-off, recoveries, bureau refreshes	expected loss, capital, downside-risk constraints
Rewards / promos	promo-state and latent response evolution	qualification, redemption, reward liability, post-promo decay	reward cost and temporary-versus-durable lift
Attrition / conversion / relationship	dormancy, closure, conversion, retention, relationship state	activity, closure, other-product observations	horizon length, relationship value, retention constraints

- a credit-card precedent for sequential NPV optimization [23];
- dynamic stochastic repayment and collections processes [2, 3];
- belief-state or POMDP control language [14, 21];
- Bayesian filtering language [4, 6, 24];
- offline evaluation and safe-improvement tools [5, 8, 11, 12, 15–18, 22].

The literature does not identify this bank’s:

- exact latent state vector,
- outside-wallet and movable-wallet laws of motion,
- reward ledger detail,
- feasible action correspondence,
- behavior-policy logging quality,

- or policy-promotion threshold.

Those objects must be justified with internal experiments, quasi-experiments, logged-action support, finance/risk reconciliation, and governance approval.

11 Implementation and governance ladder

The stochastic process implies four component interfaces: **simulate_policy** for fixed-policy forward simulation, **score_action_grid** for finite-menu line/offer/retention decisions, **evaluate_policy_offline** for comparing a candidate rule φ against the logged policy μ , and **register_policy** for versioning policy rules and their allowed-use scope. These are NPV-component interfaces, not ownership of the bank’s campaign, underwriting, line-management, or servicing platforms.

The correct rollout sequence is:

1. build the fixed-policy forward simulator under the incumbent or declared policy bundle;
2. calibrate and reconcile transitions, cash-flow compiler, and risk outputs;
3. evaluate challenger policies offline with support diagnostics, behavior- policy probabilities $\mu(a_t | H_t)$ where available, and policy-value estimators [5, 12, 22];
4. allow constrained challenger optimization only for narrow approved subproblems, preferably with discrete action grids and conservative support-aware methods [7, 8, 16, 17];
5. allow live influence only through governed experimentation and rollback controls.

This ordering is not optional. Without it, the belief-state control language can launder weakly identified modules into apparently precise policy recommendations.

12 Source-support audit

This note is a mathematical assembly of the current NPV proposal into a full state-space control specification. It is not a completed empirical validation package. Table 2 records the support class for the main claims.

Table 2: Claim-support ledger for the full stochastic-process note.

Design claim	Support class	What remains bank-specific
Credit-card customer value should be modeled as a sequential control problem.	Direct literature support plus card-management precedent	Exact state/action granularity and policy scope
Partial observability requires a belief-state or POMDP formulation.	Direct control/filtering support; proposal derivation for wallet and latent response states	Exact posterior approximation and state coordinates

Design claim	Support class	What remains bank-specific
Macro, spend, payment, loss, reward, and attrition blocks must be assembled in one stochastic system.	Proposal derivation from reused equations and literature-supported mechanisms	Exact parametrization and empirical calibration
Policy should condition on the posterior distribution, not only a point estimate.	Direct filtering and POMDP support	Chosen finite-dimensional approximation and evidence threshold
Offline policy evaluation must precede challenger optimization.	Direct OPE / safe-improvement literature support	Logged behavior-policy quality, overlap, support, and promotion rule
Safe live rollout is not implied by the literature alone.	Governance conclusion from the limits of the cited methods	All live thresholds, rollback rules, and compliance approval

13 Conclusion

A committee-ready dynamic note for the credit-card NPV component cannot stop at a high-level POMDP backbone. It must specify the whole stochastic process: macro block, usage and routing, wallet and cannibalization, balances, payments, delinquency and recovery, rewards and promotions, attrition and conversion, relationship state, observation equations, posterior update, reward mapping, and constrained policy recursion. This note does that by reassembling the current proposal’s modules into one belief-state control model.

The resulting object is mathematically strong enough to define forward simulation under a fixed policy and conceptually strong enough to support later offline policy evaluation and constrained challenger optimization. It remains properly conservative: the literature supports the control and filtering language, but the empirical state content, parameterization, and rollout thresholds remain issuer-specific obligations.

A Notation summary

Symbol	Meaning
i	customer / account / prospect / application unit
t	monthly time index
d	decision context
$a, a_0(d)$	action and baseline action
s	scenario and valuation-semantics bundle
π^{down}	downstream operating policy bundle assumed in valuation
μ	historical logged behavior policy that generated observed actions
φ	candidate or challenger decision rule
π^{gov}	governance and hard-admissibility bundle

Symbol	Meaning
O_t	observed decision state
U_t	latent state
X_t	full state (O_t, U_t)
b_t	posterior belief over the latent state
G_t	lifecycle regime
M_t^{lat}, M_t^{obs}	latent and observed macro state
W_t^{out}, W_t^{mov}	outside wallet and movable wallet
B_t	balance-subledger vector
D_t	delinquency/default/cure state
Rwd_t	reward / promotion state
Rel_t	relationship state
ΔCF_t	one-period incremental cash-flow primitive
ΔNPV	discounted incremental value

B Assumptions ledger

Assumption	Role in the model
Controlled Markov structure	Defines the latent-state law of motion under action
Observation model	Links latent/full state to observed bank signals
Belief-state sufficiency	Justifies control on posterior state rather than full history
Subledger accounting	Prevents impossible balance / revenue / loss paths
Constraint admissibility	Restricts optimization to compliant and approved actions
Offline evaluation assumptions	Required for policy-value estimation from logs
Evidence-grade separation	Prevents weak latent modules from being silently promoted to causal policy value

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